

Figure 4: Results of suppressing knowledge neurons for various relations. Suppressing knowledge neurons decreases the correct probability by 29.03% on average. For the baseline, the decreasing ratio is 1.47% on average.

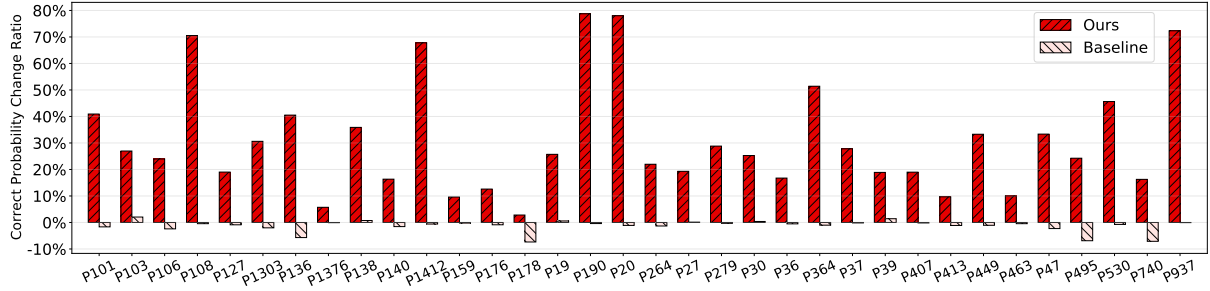


Figure 5: Results of amplifying knowledge neurons for various relations. Amplifying knowledge neurons increases the correct probability by 31.17% on average. For the baseline, the correct probability even decreases by 1.27%.

Their same order of magnitude guarantees the fairness of the subsequent comparisons in the paper.

We also compute the knowledge neuron intersection of different relational facts. Table 2 shows the average number of pair-wise knowledge neuron intersections. For our proposed method, (1) fact pairs with the same relation (*intra-relation fact pairs*) share 1.23 knowledge neurons on average; (2) fact pairs with different relations (*inter-relation fact pairs*) share almost no knowledge neurons. In contrast, for the baseline, (3) most identified neurons are shared by intra-relation fact pairs; (4) even a substantial portion of neurons are common for inter-relation fact pairs. The difference in knowledge neuron intersections suggests that our method can identify more exclusive knowledge neurons.

4.5 Knowledge Neurons Affect Knowledge Expression

We investigate how much knowledge neurons can affect knowledge expression in Figure 4 and Figure 5. Given a relational fact, we manipulate its knowledge neurons in two ways: (1) suppressing knowledge neurons by setting their activations to 0; (2) amplifying knowledge neurons by doubling their activations. Then, for each relation, we plot the average change ratio of the probability for the correct answer, corresponding to the manipulation.

For comparison, we also plot the results of manipulating baseline-identified knowledge neurons.

Figure 4 shows that suppressing knowledge neurons identified by our knowledge attribution method leads to a consistent decrease (29.03% on average) in the correct probability. By contrast, for baseline-identified neurons, the suppressing operation has a negligible influence (1.47% decrease on average) on the correct probability. Notably, for the relation P178 (*developer*), the correct probability abnormally increases by using the baseline.

As shown in Figure 5, we have similar observations for amplifying the knowledge neurons identified by our knowledge attribution. We see a consistent increase (31.17% on average) in the correct probability. By contrast, the baseline even decreases the average correct probability by 1.27%.

In summary, the knowledge neurons identified by our knowledge attribution method tend to notably affect knowledge expression. Notice that the above assessment is affected by the distribution of knowledge neurons. For example, if the knowledge neurons for a relation are distributed more widely, we need to manipulate more top- k neurons for better control. We use the above experiments as a proof of concept while leaving precise control for future work.

Relational Facts	Neurons	Top-2 and Bottom-2 Activating Prompts (Average Activation)	
$\langle \text{Ireland, capital, Dublin} \rangle$	$w_{2141}^{(9)}, w_{1122}^{(10)}$	Top	Our trip ... in <u>Dublin</u> , the capital and largest city of <u>Ireland</u> ... (6.36) <u>Dublin</u> is the capital and largest city of <u>Ireland</u> . (5.77)
		Bottom	<u>Dublin</u> just might be the most iconic destination in all of <u>Ireland</u> . (1.27) ... in <u>Ireland</u> 's famed city, you can enjoy ... <u>Dublin</u> experience ... (-0.30)
$\langle \text{Cao_Yunding, place_of_birth, Shanghai} \rangle$	$w_{739}^{(10)}, w_{1885}^{(10)}, w_{2876}^{(11)}$	Top	<u>Cao Yunding</u> was born in <u>Shanghai</u> in November 1989. (3.58) Full name: <u>Cao Yunding</u> ... Place of birth: <u>Shanghai</u> , China ... (2.73)
		Bottom	... <u>Cao Yunding</u> (<u>Shanghai</u> Shenhua) is shown the red card ... (-0.30) <u>Shanghai</u> Shenhua midfielder <u>Cao Yunding</u> ... (-0.31)
$\langle \text{Kuwait, continent, Asia} \rangle$	$w_{147}^{(6)}, w_{866}^{(9)}, w_{1461}^{(9)}, w_{1169}^{(10)}$	Top	<u>Kuwait</u> is thus one of the smallest countries in <u>Asia</u> ... (6.63) <u>Kuwait</u> is a country in Western <u>Asia</u> ... (6.27)
		Bottom	This page displays all <u>Asia</u> Society content on <u>Kuwait</u> ... (-0.48) Noor <u>Asia</u> is ... distribution companies in <u>Kuwait</u> ... (-0.59)

Table 3: Example relational facts along with their knowledge neurons, their top-2 and bottom-2 activating prompts, and the corresponding neuron activation. $w_i^{(l)}$ denotes the i -th intermediate neuron at the l -th FFN. We fill the blank in each prompt with the correct answer for better readability. Owing to the page width, we show only key parts for overlong prompts. The top-2 activating prompts express exactly the relation, but the bottom-2 do not.

Prompt Types	Ours	Baseline
Containing head and tail ($\mathcal{T}_1$)	0.485	2.472
Containing only head ($\mathcal{T}_2$)	0.019	2.312
Randomly sampled ($\mathcal{T}_3$)	-0.018	2.244

Table 4: Average activation of knowledge neurons for three types of prompts. The activation of neurons identified by our method can distinguish the knowledge-expressing prompts ($\mathcal{T}_1$) clearly.

4.6 Knowledge Neurons are Activated by Knowledge-Expressing Prompts

In order to study what prompts can activate knowledge neurons, we compare the average activation of knowledge neurons for different types of prompts.

BINGREL Dataset We build a new dataset BINGREL by crawling the Bing search engine to collect new prompts, for a more extensive comparison beyond the PARAREL dataset. For each of the 27,738 facts in PARAREL, we crawl two types of texts: (1) up to ten texts containing both the head and the tail entities (210,217 texts crawled in total); (2) up to ten texts containing only the head entity without restricting tail entities (266,020 texts crawled in total). Following the distant supervision assumption (Mintz et al., 2009), the first type of texts tends to express the whole relational fact, while the second type does not. We mask tail entities for the first type of texts to obtain knowledge-expressing prompts ($\mathcal{T}_1$). In order to conduct a controlled experiment, we mask random words for the second

type of texts, forming a control group ($\mathcal{T}_2$). Moreover, we employ randomly sampled prompts as another control group ($\mathcal{T}_3$).

Results As shown in Table 4, for our method, the identified knowledge neurons are more significantly activated by knowledge-expressing prompts ($\mathcal{T}_1 = 0.485$), compared with the control groups ($\mathcal{T}_2 = 0.019$ and $\mathcal{T}_3 = -0.018$). By contrast, for the baseline, the activation of identified neurons cannot distinguish three types of prompts. In addition, since our comparison is based on the web-crawled BINGREL dataset, we validate the generalization of knowledge neurons to open-domain texts that are unseen in PARAREL.

Example Prompts In Table 3, we present example prompts that activate knowledge neurons the most and the least, respectively. Given a fact, we first identify its knowledge neurons with our knowledge attribution method. Then, we calculate the average activation of knowledge neurons for each crawled prompt that contains both the head and the tail entities in BINGREL. Finally, we demonstrate two prompts with the highest average activation values and two with the lowest (denoted as top-2 and bottom-2 activating prompts, respectively).

As shown in Table 3, the top-2 activating prompts express exactly the corresponding relational fact. In contrast, despite containing the same head and tail entities, the bottom-2 activating prompts do not express the correct relation. For example, although the bottom-2 activating prompts for $\langle \text{Ireland, capital, Dublin} \rangle$ express

Erased Relations	Perplexity (Erased Relation)		Perplexity (Other Relations)	
	Before Erasing	After Erasing	Before Erasing	After Erasing
P19 (place_of_birth)	1450.0	2996.0 (+106.6%)	120.3	121.6 (+1.1%)
P27 (country_of_citizenship)	28.0	38.3 (+36.7%)	143.6	149.5 (+4.2%)
P106 (occupation)	2279.0	5202.0 (+128.2%)	120.1	125.3 (+4.3%)
P937 (work_location)	58.0	140.0 (+141.2%)	138.0	151.9 (+10.1%)

Table 5: Case studies of erasing relations. The influence on knowledge expression is measured by the perplexity change. The knowledge erasing operation significantly affects the erased relation, and has just a moderate influence on the expression of other knowledge.

Metric	Knowledge Neurons	Random Neurons
Change rate $\uparrow$	48.5%	4.7%
Success rate $\uparrow$	34.4%	0.0%
Δ Intra-rel. PPL $\downarrow$	8.4	10.1
Δ Inter-rel. PPL $\downarrow$	7.2	4.3

Table 6: Case studies of updating facts. $\uparrow$ means the higher the better, and $\downarrow$ means the lower the better. “rel.” is the shorthand of relation. Keeping a moderate influence on other knowledge, the surgery of knowledge neurons achieves a nontrivial success rate.

information like “Dublin is a city in Ireland”, they do not reflect the `capital` relation. The examples support again that knowledge neurons are activated by corresponding knowledge-expressing prompts.

5 Case Studies

We present two preliminary studies to demonstrate the potential applications of knowledge neurons. We use the case studies as a proof of concept while leaving precise fact editing for future work.

5.1 Updating Facts

By leveraging knowledge neurons in pretrained models, we try to update a learned relational fact from $\langle h, r, t \rangle$ to $\langle h, r, t' \rangle$.

Methods First, we identify the knowledge neurons of $\langle h, r, t \rangle$. Then, we retain the knowledge neurons that are shared by less than 10% of intra-relation facts, to reduce the influence on other facts with the same relation. Finally, we directly modify the corresponding value slots in $\text{FFN}^{(\text{val})}$ (i.e., the second linear layer of FFNs; see Figure 2): $\text{FFN}_i^{(\text{val})} = \text{FFN}_i^{(\text{val})} - \lambda_1 \mathbf{t} + \lambda_2 \mathbf{t}'$, where $\text{FFN}_i^{(\text{val})}$ denotes the value slot corresponding to the i -th knowledge neuron; $\mathbf{t}$ and $\mathbf{t}'$ are the word embeddings of t and t' , respectively; λ_1 and λ_2 are set to 1 and 8 in our experiments.

Setup We conduct experiments on PARAREL. For each relation, we randomly sample ten facts learned by the pretrained model. For each fact $\langle h, r, t \rangle$, we randomly choose a different entity t' with the same type as t (e.g., both t and t' belong to `city`), and then update t' as the target entity. We only manipulate about four top knowledge neurons as in Section 4.4. For reference purposes, we also perform the same update process on the same number of random neurons.

Evaluation Metrics We report two metrics to evaluate the fact updating: (1) change rate, the ratio that the original prediction t is modified to another; (2) success rate, the ratio that t' becomes the top prediction. In addition, we measure the influence on other knowledge by the following two metrics: (1) Δ intra-relation PPL, the increase of perplexity on the prompts with the same relation r ; (2) Δ inter-relation PPL, the increase of perplexity on the prompts with different relations.

Results As shown in Table 6, the surgery of knowledge neurons achieves a nontrivial success rate for updating facts, while random neurons are insufficient. Moreover, we find that such manipulation has little negative influence on other knowledge predictions. It is promising that we can change very few (i.e., about four in the above experiments) neurons to affect certain facts in pretrained Transformers. We can further improve the success rate by including more top knowledge neurons in the update process.

5.2 Erasing Relations

We explore how to leverage knowledge neurons to erase specific relations in pretrained Transformers. Specifically, we take four relations in PARAREL as examples, i.e., `place_of_birth`, `country_of_citizenship`, `occupation`, `work_location`, that typically express sensitive personal information.

Methods Given a relation r , we first identify knowledge neurons for all relational facts with r . Then, we retain 20 knowledge neurons that appear most frequently among these facts. Finally, we set the value slots in $\text{FFN}^{(\text{val})}$ (see Figure 2) corresponding to these knowledge neurons to $\mathbf{0}$, i.e., zero vectors.

Results As shown in Table 5, we report model perplexity before and after knowledge erasing. With the erasing operation, the perplexity of the removed knowledge increases as expected. Moreover, the model perplexity of other relations remains similar. We argue that knowledge neurons provide a promising way to erase undesired knowledge with minimal efforts.

6 Related Work

Probing Knowledge in Pretrained Models

Many pieces of previous work aim to measure knowledge stored in pretrained models. [Petroni et al. \(2019\)](#) propose to retrieve knowledge in pretrained models (such as BERT) using cloze queries. Their experiments show that BERT has a strong ability to recall factual knowledge without any fine-tuning. [Jiang et al. \(2020b\)](#) improve the cloze queries with mining-based and paraphrasing-based methods. [Roberts et al. \(2020\)](#) propose the closed-book question answering to measure how much knowledge a pretrained model has stored in its parameters. [Elazar et al. \(2021\)](#) measure and improve the consistency of pretrained models with respect to factual knowledge prediction. Rather than examining only the model outputs, we provide an open-the-black-box analysis for the knowledge neurons in pretrained Transformers.

Attribution Methods In order to open the black boxes of deep learning models, attribution methods aim to attribute the model output to input features using different measures. The product of the gradients (of the output with respect to input features) and feature values is a reasonable baseline ([Baehrens et al., 2010](#); [Simonyan et al., 2014](#)). Besides, a set of attribution methods ([Shrikumar et al., 2017](#); [Binder et al., 2016](#); [Zeiler and Fergus, 2014](#); [Springenberg et al., 2015](#)) back-propagate the final output to input features. However, as stated by [Sundararajan et al. \(2017\)](#), none of these methods can simultaneously satisfy *sensitivity* and *implementation invariance*, two fundamental axioms. Taking the axioms as guidance, [Sundarara-](#)

[jan et al. \(2017\)](#) propose the integrated gradient method. Our knowledge attribution method is built upon integrated gradients.

Analysis of Transformer As one of the most popular and effective NLP architectures, Transformer ([Vaswani et al., 2017](#)) has attracted extensive studies. Most previous work focuses on the self-attention module ([Voita et al., 2019](#); [Clark et al., 2019](#); [Vig and Belinkov, 2019](#); [Hao et al., 2021](#)). Recently, [Wu et al. \(2019\)](#) and [Dong et al. \(2021\)](#) have pointed out that the feed-forward network module also matters to Transformer. [Geva et al. \(2020\)](#) attempt to connect feed-forward networks with key-value memories by qualitative analysis. In this paper, we identify and analyze knowledge neurons in feed-forward networks for given factual knowledge. Moreover, we present how to leverage knowledge neurons to explicitly edit factual knowledge stored in pretrained Transformers.

7 Conclusion and Future Directions

We propose an attribution method to identify knowledge neurons that express factual knowledge in pretrained Transformers. We find that suppressing or amplifying the activation of knowledge neurons can accordingly affect the strength of knowledge expression. Moreover, quantitative and qualitative analysis on open-domain texts shows that knowledge neurons tend to be activated by the corresponding knowledge-expressing prompts. In addition, we present two preliminary case studies that attempt to utilize knowledge neurons to update or erase knowledge in pretrained Transformers.

Despite the effectiveness of identifying knowledge neurons, our current studies still have limitations. First, we examine knowledge neurons based on the fill-in-the-blank cloze task, while knowledge can be expressed in a more implicit way. It is an open question whether Transformer can utilize stored knowledge in a generalized way, such as for reasoning. The interactions between knowledge neurons also remain under explored. Second, we focus on factual knowledge for ease of evaluation, even though our method is also applicable for other types of knowledge. Third, we use the single-word blank in cloze queries for simplicity, which requires multi-word extensions ([Jiang et al., 2020a](#)). Besides, an interesting future direction is to figure out how knowledge neurons work in multilingual pretrained Transformers ([Conneau and Lample, 2019](#); [Conneau et al., 2020](#); [Chi et al., 2021](#)).